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LT-(1)

Module IV

Laplace Transform

Let $f(t)$ be a given function that is defined for all $t \geq 0$. Then

$$L[f(t)] = F(s) = \int_0^{\infty} e^{-st} f(t) dt \text{ is called}$$

the Laplace transform of $f(t)$.

Note:- Here the original function depends on t and its transform on s . That is, a function in one space is transformed to a function in another space. or time domain is changed to frequency domain.

The original function $f(t)$ is called inverse of $F(s)$ denoted by $f(t) = L^{-1}[F(s)]$

LT-(2)

Laplace transform of some important function

$f(t)$	$F(s) = L[f(t)]$
1) 0	0
2) 1	$\frac{1}{s}$
3) t	$\frac{1}{s^2}$
4) e^{at}	$\frac{1}{s-a}$
5) $t^n, n \geq 0$	$\frac{n!}{s^{n+1}}$
6) $\cos at$	$\frac{s}{s^2+a^2}$
7) $\sin at$	$\frac{a}{s^2+a^2}$
8) $\cosh at$	$\frac{s}{s^2-a^2}$
9) $\sinh at$	$\frac{a}{s^2-a^2}$
10) $e^{at} \sin bt$	$\frac{b}{(s-a)^2+b^2}$
11) $e^{at} \cos bt$	$\frac{(s-a)}{(s-a)^2+b^2}$
12) $e^{at} t^n$	$\frac{n!}{(s-a)^{n+1}}$

LT-(3)

$$1) L(0) = \int_0^{\infty} e^{-st}(0) dt = \underline{\underline{0}}$$

$$2) L(1) = \int_0^{\infty} e^{-st}(1) dt = \left[\frac{e^{-st}}{-s} \right]_0^{\infty} = \underline{\underline{\frac{1}{s}}}$$

$$3) L(t) = \int_0^{\infty} e^{-st}(t) dt = \left[t \left(\frac{e^{-st}}{-s} \right) - \frac{e^{-st}}{s^2} \right]_0^{\infty} = \underline{\underline{\frac{1}{s^2}}}$$

$$4) L(e^{at}) = \int_0^{\infty} e^{-st} e^{at} dt = \int_0^{\infty} e^{-(s-a)t} dt = \left[\frac{e^{-(s-a)t}}{-(s-a)} \right]_0^{\infty} = \underline{\underline{\frac{1}{s-a}}}$$

Linearity Property

$$L[af(t) + bg(t)] = aL[f(t)] + bL[g(t)]$$

$$8 \& 9) L(\cosh at) \& L(\sinh at)$$

$$\cosh at = \frac{e^{at} + e^{-at}}{2}$$

$$\sinh at = \frac{e^{at} - e^{-at}}{2}$$

$$L(\cosh at) = L\left[\frac{e^{at} + e^{-at}}{2}\right] = \frac{1}{2}L(e^{at}) + \frac{1}{2}L(e^{-at})$$

$$= \frac{1}{2} \left[\frac{1}{s-a} + \frac{1}{s+a} \right] = \underline{\underline{\frac{s}{s^2 - a^2}}}$$

$$L(\sinh at) = L\left[\frac{e^{at} - e^{-at}}{2}\right] = \frac{1}{2} \left[\frac{1}{s-a} - \frac{1}{s+a} \right] = \underline{\underline{\frac{a}{s^2 - a^2}}}$$

$$6 \& 7) L(\cos at) \& L(\sin at)$$

$$L(e^{iat}) = L(\cos at) + iL(\sin at)$$

$$[\because e^{iat} = \cos at + i \sin at]$$

$$\Rightarrow \frac{1}{s-ia} = L(\cos at) + iL(\sin at)$$

LT- (4)

$$\Rightarrow \frac{s+ia}{s^2+a^2} = L(\cos at) + iL(\sin at)$$

Comparing real & imaginary parts

$$L(\cos at) = \frac{s}{s^2+a^2} \quad \& \quad L(\sin at) = \frac{a}{s^2+a^2}$$

$$\begin{aligned} 5) \quad L(t^n) &= \int_0^{\infty} e^{-st} t^n dt = \left(t^n \left(\frac{e^{-st}}{-s} \right) \right)_0^{\infty} + \int_0^{\infty} \frac{e^{-st}}{s} n t^{n-1} dt \\ &= \frac{n}{s} \int_0^{\infty} e^{-st} t^{n-1} dt \\ &= \frac{n}{s} L(t^{n-1}) \quad \text{--- (1)} \end{aligned}$$

Similarly,

$$L(t^{n-1}) = \frac{n-1}{s} L(t^{n-2})$$

$$L(t^{n-2}) = \frac{n-2}{s} L(t^{n-3})$$

$$\vdots$$

$$L(t^2) = \frac{2}{s} L(t)$$

$$L(t) = \frac{1}{s} L(1)$$

$$L(1) = \frac{1}{s}$$

Substituting in (1)

$$L(t^n) = \frac{n}{s} \left(\frac{n-1}{s} \right) \left(\frac{n-2}{s} \right) \dots \left(\frac{2}{s} \right) \left(\frac{1}{s} \right) \left(\frac{1}{s} \right)$$

$$= \frac{n!}{s^{n+1}}$$

LT-(5)

First shifting theorem

If $L[f(t)] = F(s)$, then $L[e^{at} f(t)] = F(s-a)$

10) $L(e^{at} \sin bt)$

Let $f(t) = \sin bt \Rightarrow F(s) = L(\sin bt) = \frac{b}{s^2 + b^2}$

$$\therefore L(e^{at} \sin bt) = F(s-a) = \frac{b}{(s-a)^2 + b^2}$$

11) $L(e^{at} \cos bt)$

$f(t) = \cos bt \Rightarrow F(s) = L(\cos bt) = \frac{s}{s^2 + b^2}$

$$L(e^{at} \cos bt) = F(s-a) = \frac{(s-a)}{(s-a)^2 + b^2}$$

12) $L(e^{at} t^n)$

$f(t) = t^n \Rightarrow F(s) = L(t^n) = \frac{n!}{s^{n+1}}$

$$\therefore L(e^{at} t^n) = F(s-a) = \frac{n!}{(s-a)^{n+1}}$$

Problems

1) $L(2t+8) = 2L(t) + 8L(1) = \frac{2}{s^2} + \frac{8}{s}$

2) $L(a-bt)^2 = L(a^2 + b^2 t^2 - 2abt) = \frac{a^2}{s} + b^2 \left(\frac{2!}{s^3} \right) - \frac{2ab}{s^2}$

3) $L(\cos 2\pi t) = \frac{s}{s^2 + 4\pi^2}$

LT-(6)

$$4) \quad L(\cos^2 2t) = L\left[\frac{1 + \cos 4t}{2}\right] = \frac{1}{2} \left[\frac{1}{s} + \frac{s}{s^2 + 16} \right]$$

$$\begin{aligned} 5) \quad L(\sin 3t \cos 2t) &= L\left[\frac{\sin 5t + \sin t}{2}\right] \\ &= \frac{1}{2} \left[\frac{5}{s^2 + 25} + \frac{1}{s^2 + 1} \right] \end{aligned}$$

$$\begin{aligned} 6) \quad L(e^{3t} \sinh 4t) \\ \text{Let } f(t) = \sinh 4t \Rightarrow F(s) = L(\sinh 4t) \\ = \frac{4}{s^2 - 16} \end{aligned}$$

$$\therefore L(e^{3t} \sinh 4t) = F(s-3) = \frac{4}{(s-3)^2 - 16}$$

$$\begin{aligned} 7) \quad L(\cos(3t+\theta)) &= L[\cos 3t \cos \theta - \sin 3t \sin \theta] \\ &= \cos \theta L(\cos 3t) - \sin \theta L(\sin 3t) \\ &= \cos \theta \left(\frac{s}{s^2 + 9} \right) - \sin \theta \left(\frac{3}{s^2 + 9} \right) \end{aligned}$$

$$\begin{aligned} 8) \quad L\left[2 \sin\left(3t - \frac{\pi}{2}\right)\right] &= 2 L\left[-\sin\left(\frac{\pi}{2} - 3t\right)\right] = -2 L(\cos 3t) \\ &= \frac{-2s}{s^2 + 9} \end{aligned}$$

$$\begin{aligned} 9) \quad L(t^3 e^{-2t}) \\ \text{Let } f(t) = t^3 \Rightarrow F(s) = L(t^3) = \frac{3!}{s^4} \\ \therefore L(e^{-2t} t^3) = F(s+2) = \frac{3!}{(s+2)^4} \end{aligned}$$

LT-(7)

$$10) \quad L \left(2 e^{-\frac{t}{2}} \sin 4\pi t \right)$$

$$f(t) = \sin 4\pi t \Rightarrow F(s) = L(\sin 4\pi t) = \frac{4\pi}{s^2 + 16\pi^2}$$

$$\therefore L \left(2 e^{-\frac{t}{2}} \sin 4\pi t \right) = 2 F \left(s + \frac{1}{2} \right) = 2 \left[\frac{4\pi}{\left(s + \frac{1}{2} \right)^2 + 16\pi^2} \right]$$

$$11) \quad L(\sinh t \cos t) = L \left[\left(\frac{e^t - e^{-t}}{2} \right) \cos t \right]$$

$$= \frac{1}{2} \left[L(e^t \cos t) - L(e^{-t} \cos t) \right]$$

$$= \frac{1}{2} \left[\frac{s-1}{(s-1)^2 + 1} - \frac{s+1}{(s+1)^2 + 1} \right]$$

$$12) \quad L(\cos^3 2t) = L \left(\frac{\cos 6t + 3 \cos 2t}{4} \right)$$

$$= \frac{1}{4} \left[\frac{s}{s^2 + 36} + \frac{3s}{s^2 + 4} \right]$$

Result : If $L[f(t)] = F(s)$, then
 $L[t^n f(t)] = (-1)^n F^{(n)}(s)$

$$1) \quad L(t \sin 5t)$$

Soln : Let $f(t) = \sin 5t \Rightarrow F(s) = L(\sin 5t) = \frac{5}{s^2 + 25}$

$$\therefore L(t \sin 5t) = (-1)^1 F'(s) = \frac{10s}{(s^2 + 25)^2}$$

LT-(8)

2) $L(t \cos 3t)$

Soln:- Let $f(t) = \cos 3t \Rightarrow F(s) = L(\cos 3t) = \frac{s}{s^2 + 9}$

$$\therefore L(t \cos 3t) = (-1)' F'(s) = \underline{\underline{\frac{s^2 - 9}{(s^2 + 9)^2}}}$$

3) $L(t^2 \sinh at)$

Soln:- Let $f(t) = \sinh at \Rightarrow F(s) = L(\sinh at) = \frac{a}{s^2 - a^2}$

$$\therefore L(t^2 \sinh at) = (-1)'' F''(s)$$

$$= F''\left(\frac{-4s}{(s^2 - 4)^2}\right)$$

$$= \frac{(s^2 - 4)^2(-4) + 4s(2)(s^2 - 4)(2s)}{(s^2 - 4)^4}$$

$$= \underline{\underline{\frac{12s^2 + 16}{(s^2 - 4)^3}}}$$

Result:- If $L(f(t)) = F(s)$, then $L\left(\frac{f(t)}{t}\right) = \int_s^\infty F(s) ds$

1. $L\left(\frac{\sin t}{t}\right)$

Soln:- Let $f(t) = \sin t \Rightarrow F(s) = L(\sin t) = \frac{1}{s^2 + 1}$

$$\therefore L\left(\frac{\sin t}{t}\right) = \int_s^\infty \left(\frac{1}{s^2 + 1}\right) ds = \left[\tan^{-1} s\right]_s^\infty = \underline{\underline{\frac{\pi}{2} - \tan^{-1} s}}$$

LT-(9)

$$2) L\left(\frac{\sin^2 t}{t}\right) = L\left[\frac{1 - \cos 2t}{2t}\right]$$

$$\text{Let } f(t) = 1 - \cos 2t \Rightarrow F(s) = L(1 - \cos 2t) \\ = \frac{1}{s} - \frac{s}{s^2 + 4}$$

$$\begin{aligned} \therefore L\left[\frac{1 - \cos 2t}{2t}\right] &= \frac{1}{2} \int_s^\infty \left(\frac{1}{s} - \frac{s}{s^2 + 4}\right) ds \\ &= \frac{1}{2} \left[\log s - \frac{1}{2} \log(s^2 + 4) \right]_s^\infty \\ &= \frac{1}{2} \left[\log \frac{s}{\sqrt{s^2 + 4}} \right]_s^\infty \\ &= \frac{1}{2} \left[\log \left(\frac{1}{\sqrt{1 + \frac{4}{s^2}}} \right) \right]_s^\infty \\ &= \frac{1}{2} \left[\log 1 - \log \sqrt{1 + \frac{4}{s^2}} \right]_s^\infty \\ &= \frac{1}{2} \log \sqrt{1 + \frac{4}{s^2}} = \underline{\underline{\frac{1}{2} \log \left(\frac{\sqrt{s^2 + 4}}{s} \right)}} \end{aligned}$$

$$3) L\left[\frac{\cos 3t - \cos 2t}{t}\right]$$

$$\text{Soln: Let } f(t) = \cos 3t - \cos 2t \Rightarrow F(s) = \frac{s}{s^2 + 9} - \frac{s}{s^2 + 4}$$

$$\begin{aligned} \therefore L\left[\frac{\cos 3t - \cos 2t}{t}\right] &= \int_s^\infty \left(\frac{s}{s^2 + 9} - \frac{s}{s^2 + 4}\right) ds \\ &= \frac{1}{2} \left[\log(s^2 + 9) - \log(s^2 + 4) \right]_s^\infty \\ &= \frac{1}{2} \left[\log \left(\frac{s^2 + 9}{s^2 + 4} \right) \right]_s^\infty \\ &= \frac{1}{2} \left[\log \left(\frac{1 + \frac{9}{s^2}}{1 + \frac{4}{s^2}} \right) \right]_s^\infty \\ &= \underline{\underline{\frac{1}{2} \log \left(\frac{s^2 + 9}{s^2 + 4} \right)}} \end{aligned}$$

Problems

Find the Laplace transform of

1) $\sin at \cos bt$

2) $\sin^3 2t$

3) $e^{-3t} (\cos 4t + 3 \sin 4t)$

4) $t e^{-4t} \sin 3t$

5) $\cos(at+b)$

6) $\sin at \sin bt$

7) $\sin^2 3t$

8) $e^{-2t} \sin 4t$

9) $t \sin^2 3t$

10) $t \sin 3t \cos 2t$

11) $t^2 e^{-2t} \cos t$

12) $\frac{e^{-t} \sin t}{t}$

13) $\frac{1 - \cos 2t}{t}$

LT- (10)

Inverse Laplace Transforms

$F(s)$	$f(t) = L^{-1}[F(s)]$
1) 0	0
2) $\frac{1}{s}$	1
3) $\frac{1}{s^2}$	t
4) $\frac{1}{s-a}$	e^{at}
5) $\frac{1}{s^{n+1}}$	$\frac{t^n}{n!}$
6) $\frac{1}{s^2+a^2}$	$\frac{1}{a} \sin at$
7) $\frac{s}{s^2+a^2}$	$\cos at$
8) $\frac{1}{s^2-a^2}$	$\frac{1}{a} \sinh at$
9) $\frac{s}{s^2-a^2}$	$\cosh at$
10) $\frac{1}{(s-a)^2 + b^2}$	$\frac{1}{b} e^{at} \sin bt$
11) $\frac{(s-a)}{(s-a)^2 + b^2}$	$e^{at} \cos bt$
12) $\frac{1}{(s-a)^{n+1}}$	$e^{at} \frac{t^n}{n!}$

LT-(11)

1) Find $\mathcal{L}^{-1} \left[\frac{2s+14}{s^2+196} \right]$

Soln: $\mathcal{L}^{-1} \left[\frac{2s+14}{s^2+196} \right] = 2 \mathcal{L}^{-1} \left[\frac{s}{s^2+(14)^2} \right] + \mathcal{L}^{-1} \left[\frac{14}{s^2+(14)^2} \right]$
 $= 2 \cos 14t + \sin 14t$

2) $\mathcal{L}^{-1} \left(\frac{s}{\rho^2 s^2 + \frac{\pi^2}{4}} \right)$

Soln: $\mathcal{L}^{-1} \left[\frac{s}{\rho^2 s^2 + \frac{\pi^2}{4}} \right] = \frac{1}{\rho^2} \mathcal{L}^{-1} \left[\frac{s}{s^2 + \frac{\pi^2}{4\rho^2}} \right] = \frac{1}{\rho^2} \cos \left(\frac{\pi}{2\rho} t \right)$

3) Find $\mathcal{L}^{-1} \left[\frac{2}{s^4} - \frac{48}{s^6} \right]$

Soln: $\mathcal{L}^{-1} \left[\frac{2}{s^4} - \frac{48}{s^6} \right] = 2 \mathcal{L}^{-1} \left[\frac{1}{s^4} \right] - 48 \mathcal{L}^{-1} \left[\frac{1}{s^6} \right]$
 $= 2 \frac{t^3}{3!} - 48 \frac{t^5}{5!}$
 $= \frac{t^3}{3} - \frac{2t^5}{5}$

4) Find $\mathcal{L}^{-1} \left[\frac{1}{(s(s+1)(s+2))} \right]$

Soln: $\frac{1}{s(s+1)(s+2)} = \frac{A}{s} + \frac{B}{s+1} + \frac{C}{s+2}$

$\Rightarrow 1 = A(s+1)(s+2) + Bs(s+2) + Cs(s+1)$
 Put $s=0 \Rightarrow 1 = 2A \Rightarrow A = \frac{1}{2}$; Put $s=-1 \Rightarrow 1 = -B \Rightarrow B = -1$
 Put $s=-2 \Rightarrow 1 = 2C \Rightarrow C = \frac{1}{2}$

LT-(12)

$$\therefore \mathcal{L}^{-1}\left[\frac{1}{s(s+1)(s+2)}\right] = \frac{1}{2} \mathcal{L}^{-1}\left(\frac{1}{s}\right) - \mathcal{L}^{-1}\left(\frac{1}{s+1}\right) + \frac{1}{2} \mathcal{L}^{-1}\left(\frac{1}{s+2}\right)$$

$$= \frac{1}{2} - \underline{\underline{e^{-t}}} + \frac{1}{2} e^{-2t}$$

5) Find $\mathcal{L}^{-1}\left[\frac{1}{s^2(s+1)}\right]$

Soln :- $\frac{1}{s^2(s+1)} = \frac{A}{s} + \frac{B}{s^2} + \frac{C}{s+1}$

$$\Rightarrow 1 = As(s+1) + B(s+1) + Cs^2$$

Put $s=0 \Rightarrow 1 = B$

Put $s=-1 \Rightarrow 1 = C$

Compare the coefficient of s^2
 $0 = A + C \Rightarrow 0 = A + 1 \Rightarrow A = -1$

$$\therefore \mathcal{L}^{-1}\left[\frac{1}{s^2(s+1)}\right] = \mathcal{L}^{-1}\left(\frac{-1}{s}\right) + \mathcal{L}^{-1}\left(\frac{1}{s^2}\right) + \mathcal{L}^{-1}\left(\frac{1}{s+1}\right)$$

$$= \underline{\underline{-1 + t + e^{-t}}}$$

6) Find $\mathcal{L}^{-1}\left[\frac{-s+11}{s^2-2s-3}\right]$

Soln :- $\frac{-s+11}{(s-3)(s+1)} = \frac{2}{s-3} + \frac{(-3)}{s+1}$ (Using partial fractions)

$$\therefore \mathcal{L}^{-1}\left[\frac{-s+11}{s^2-2s-3}\right] = 2\mathcal{L}^{-1}\left[\frac{1}{s-3}\right] - 3\mathcal{L}^{-1}\left[\frac{1}{s+1}\right] = \underline{\underline{2e^{3t} - 3e^{-t}}}$$

OR (another method)

$$\mathcal{L}^{-1}\left[\frac{-s+11}{s^2-2s-3}\right] = \mathcal{L}^{-1}\left[\frac{-s+11}{(s-1)^2-4}\right] = \mathcal{L}^{-1}\left[\frac{-(s-1)+10}{(s-1)^2-4}\right]$$

$$\mathcal{L}^{-1}(3)$$

$$= -\mathcal{L}^{-1}\left[\frac{(s-1)}{(s-1)^2 - 2^2}\right] + 10\mathcal{L}^{-1}\left[\frac{1}{(s-1)^2 - 2^2}\right]$$

$$= -e^t \cosh 2t + \frac{10}{2} e^t \sinh 2t$$

7) Find $\mathcal{L}^{-1}\left[\frac{2\pi}{(s+\pi)^3}\right]$

Soln : $\mathcal{L}^{-1}\left[\frac{2\pi}{(s+\pi)^3}\right] = 2\pi e^{-\pi t} \frac{t^2}{2!} = \underline{\underline{\pi e^{-\pi t} t^2}}$

8) Find $\mathcal{L}^{-1}\left[\frac{2}{s^2 + 8s + 12}\right]$

Soln : $\mathcal{L}^{-1}\left[\frac{2}{s^2 + 8s + 12}\right] = \mathcal{L}^{-1}\left[\frac{2}{(s+4)^2 - 4}\right]$

$$= e^{-4t} \underline{\underline{\sinh 2t}}$$

9) Find $\mathcal{L}^{-1}\left[\frac{6s+7}{2s^2 + 4s + 10}\right]$

Soln : $\mathcal{L}^{-1}\left[\frac{6s+7}{2s^2 + 4s + 10}\right] = \mathcal{L}^{-1}\left[\frac{6s+7}{2(s^2 + 2s + 5)}\right]$

$$= \frac{1}{2} \mathcal{L}^{-1}\left[\frac{6s+7}{(s+1)^2 + 4}\right]$$

$$= \frac{1}{2} \mathcal{L}^{-1}\left[\frac{6(s+1)+1}{(s+1)^2 + 4}\right]$$

$$= \frac{1}{2} \left\{ \mathcal{L}^{-1}\left[\frac{6(s+1)}{(s+1)^2 + 4}\right] + \mathcal{L}^{-1}\left[\frac{1}{(s+1)^2 + 4}\right] \right\}$$

$$= \frac{1}{2} \left[\underline{\underline{6e^{-t} \cos 2t}} + \frac{1}{2} e^{-t} \sin 2t \right]$$

LT- (14)

10) Find $\mathcal{L}^{-1} \left[\frac{3s - 137}{s^2 + 2s + 401} \right]$

Soln: $\mathcal{L}^{-1} \left[\frac{3s - 137}{s^2 + 2s + 401} \right] = \mathcal{L}^{-1} \left[\frac{3s - 137}{(s+1)^2 + 400} \right]$

$$= \mathcal{L}^{-1} \left[\frac{3(s+1) - 140}{(s+1)^2 + 400} \right]$$

$$= 3e^{-t} \cos 20t - \frac{140}{20} e^{-t} \sin 20t$$

$$= \underline{\underline{3e^{-t} \cos 20t - 7e^{-t} \sin 20t}}$$

H.w

1) Find

i) $\mathcal{L}^{-1} \left[\frac{5s+1}{s^2-25} \right]$

iv) $\mathcal{L}^{-1} \left[\frac{6}{(s+1)^3} \right]$

ii) $\mathcal{L}^{-1} \left[\frac{1}{(s+\sqrt{2})(s-\sqrt{3})} \right]$

v) $\mathcal{L}^{-1} \left[\frac{90}{(s+\sqrt{3})^4} \right]$

iii) $\mathcal{L}^{-1} \left[\frac{4s+32}{s^2-16} \right]$

vi) $\mathcal{L}^{-1} \left[\frac{1}{s^3 - 2\pi s^2} \right]$

LT(1) - (15)

Convolution theorem

Convolution :- (mathematical way of combining two signals to form a third signal)

The convolution of two functions $f(t)$ and $g(t)$ denoted by $f * g(t)$ is defined as $\int_0^t f(u)g(t-u)du$

Convolution theorem

Let $L[f(t)] = F(s)$ and $L[g(t)] = G(s)$.

Then $L(f * g) = L[f(t)] L[g(t)] = F(s) G(s)$

$$\Rightarrow f * g = L^{-1}[F(s) G(s)] = \int_0^t f(u)g(t-u)du$$

1) Find $1 * (-1)$

Soln :- $f(t) = 1$ $g(t) = -1$

$$f * g = 1 * (-1) = \int_0^t 1(-1)du = -[u]_0^t = \underline{\underline{-t}}$$

2) Find $e^{-t} * e^t$

$$\begin{aligned} \text{Soln :- } e^{-t} * e^t &= \int_0^t f(u)g(t-u)du = \int_0^t e^{-u} e^{t-u} du \\ &= \int_0^t e^{t-2u} du = \left[\frac{e^{t-2u}}{-2} \right]_0^t \\ &= \frac{e^t - e^{-t}}{2} = \underline{\underline{\sinh t}} \end{aligned}$$

3) Find $\cos wt * 1$

$$\text{Soln :- } \cos wt * 1 = \int_0^t \cos wu du = \left[\frac{\sin wu}{w} \right]_0^t = \underline{\underline{\frac{\sin wt}{w}}}$$

4) Find $t * e^{-t}$

$$\begin{aligned} \text{Soln :- } t * e^{-t} &= \int_0^t u e^{-(t-u)} du = \left[u e^{-(t-u)} - e^{-(t-u)} \right]_0^t \\ &= \underline{\underline{t-1+e^{-t}}} \end{aligned}$$

(2) LT- (16)

H.W

Find i) $1 + \sin wt$

ii) $\cos wt + \cos wt$

iii) $e^{at} + e^{bt}$ ($a \neq b$)

To find inverse using convolution theorem

i) Find $L^{-1}\left(\frac{1}{(s-a)s}\right)$

Soln: Let $F(s) = \frac{1}{s-a}$

$$G(s) = \frac{1}{s}$$

$$\Rightarrow f(t) = L^{-1}[F(s)] = L^{-1}\left(\frac{1}{s-a}\right) = e^{at}$$

$$g(t) = L^{-1}[G(s)] = L^{-1}\left(\frac{1}{s}\right) = 1$$

$$\therefore L^{-1}\left(\frac{1}{(s-a)s}\right) = \int_0^t f(u)g(t-u)du = \int_0^t e^{au}(1)du = \left[\frac{e^{au}}{a}\right]_0^t = \frac{e^{at} - 1}{a}$$

a) Find $L^{-1}\left(\frac{1}{(s+1)(s-4)}\right)$

Soln: Let $F(s) = \frac{1}{s+1}$

$$G(s) = \frac{1}{s-4}$$

$$\Rightarrow f(t) = L^{-1}\left(\frac{1}{s+1}\right) = e^{-t}$$

$$\Rightarrow g(t) = L^{-1}\left(\frac{1}{s-4}\right) = e^{4t}$$

$$L^{-1}\left(\frac{1}{(s+1)(s-4)}\right) = \int_0^t e^{-u} e^{4(t-u)} du = \int_0^t e^{4t-5u} du = \left[\frac{e^{4t-5u}}{-5}\right]_0^t = \frac{e^{-t} - e^{4t}}{-5}$$

Q) L7 - (17)

3) Find $\mathcal{L}^{-1}\left[\frac{1}{(s^2 + \omega^2)^2}\right]$

Soln:- $F(s) = \frac{1}{s^2 + \omega^2}$

$G(s) = \frac{1}{s^2 + \omega^2}$

$f(t) = \mathcal{L}^{-1}\left[\frac{1}{s^2 + \omega^2}\right] = \frac{\sin \omega t}{\omega}$

$g(t) = \mathcal{L}^{-1}\left[\frac{1}{s^2 + \omega^2}\right] = \frac{\sin \omega t}{\omega}$

$$\begin{aligned} \therefore \mathcal{L}^{-1}\left[\frac{1}{(s^2 + \omega^2)^2}\right] &= \int_0^t \frac{\sin \omega u}{\omega} \cdot \frac{\sin \omega (t-u)}{\omega} du \\ &= \frac{1}{2\omega^2} \int_0^t [\cos(2\omega u - \omega t) - \cos \omega t] du \\ &= \frac{1}{2\omega^2} \left[\frac{\sin(2\omega u - \omega t)}{2\omega} - u \cos \omega t \right]_0^t \\ &= \frac{1}{2\omega^2} \left[\frac{\sin \omega t}{2\omega} - t \cos \omega t + \frac{\sin \omega t}{2\omega} \right] \\ &= \frac{1}{2\omega^2} \left[\frac{\sin \omega t}{\omega} - t \cos \omega t \right] \end{aligned}$$

4) Find $\mathcal{L}^{-1}\left[\frac{18s}{(s^2 + 36)^2}\right]$

Soln:- $F(s) = \frac{s}{s^2 + 36}$

$G(s) = \frac{1}{s^2 + 36}$

$f(t) = \mathcal{L}^{-1}\left[\frac{s}{s^2 + 36}\right] = \cos 6t$

$g(t) = \mathcal{L}^{-1}\left[\frac{1}{s^2 + 36}\right] = \frac{\sin 6t}{6}$

$$\begin{aligned} \therefore \mathcal{L}^{-1}\left[\frac{18s}{(s^2 + 36)^2}\right] &= 18 \int_0^t \cos 6u \cdot \frac{\sin 6(t-u)}{6} du \\ &= \frac{3}{2} \int_0^t \sin 6t + \sin(6t - 12u) du \\ &= \frac{3}{2} \left[u \sin 6t + \frac{\cos(6t - 12u)}{12} \right]_0^t = \underline{\underline{\frac{3}{2} t \sin 6t}} \end{aligned}$$

(4) LT - (18)

H.W

Find the inverse using convolution theorem.

i) $\frac{1}{(s-a)^2}$

ii) $\frac{2\pi s}{(s^2 + \pi^2)^2}$

iii) $\frac{9}{s(s+3)}$

iv) $\frac{\omega}{s^2(s^2 + \omega^2)}$

v) $\frac{40}{s(s^2 - 9)}$

vi) $\frac{1}{(s^2 + 1)(s^2 + 25)}$

(S) L7 - (19)

Laplace transform of derivatives & integrals

$$L[f'(t)] = s L[f(t)] - f(0)$$

$$L[f''(t)] = s^2 L[f(t)] - s f(0) - f'(0)$$

$$L[f'''(t)] = s^3 L[f(t)] - s^2 f(0) - s f'(0) - f''(0)$$

$$\therefore L[f^{(n)}(t)] = s^n L[f(t)] - s^{n-1} f(0) - s^{n-2} f'(0) - \dots - f^{(n-1)}(0)$$

1. Using the formula for Laplace transform of derivatives, find $L(\sin at)$.

Soln: - $f(t) = \sin at$ $f'(t) = a \cos at$ $f''(t) = -a \sin at$

$$L[f''(t)] = s^2 L[f(t)] - s f(0) - f'(0)$$

$$\Rightarrow L[-a \sin at] = s^2 L[\sin at] - a$$

$$\Rightarrow L(\sin at)(s^2 + a^2) = a$$

$$\Rightarrow L(\sin at) = \frac{a}{s^2 + a^2}$$

2. $L(te^{at})$

$$f(t) = te^{at} \quad f'(t) = ate^{at} + e^{at}$$

$$L[f'(t)] = s L[f(t)] - f(0)$$

$$L[ate^{at} + e^{at}] = s L(te^{at})$$

$$\Rightarrow L(te^{at})(s - a) = L(e^{at}) = \frac{1}{s - a}$$

$$\Rightarrow L(te^{at}) = \frac{1}{(s - a)^2}$$

(6) LT- QO)

H.W

i) $f(t) = t \sin at$

ii) $f(t) = t \cos at$

iii) $f(t) = \cos^2 \omega t$

Laplace transform of integrals

(w) 7- y $L[f(t)] = F(s)$ then $L\left[\int_0^t f(\tau) d\tau\right] = \frac{F(s)}{s}$

Thus $L^{-1}\left[\frac{F(s)}{s}\right] = \int_0^t f(\tau) d\tau$

1. Find the inverse using integration.

i) $\frac{1}{s(s^2 + \omega^2)}$

Soln: Let $F(s) = \frac{1}{s(s^2 + \omega^2)} \Rightarrow f(t) = L^{-1}\left(\frac{1}{s(s^2 + \omega^2)}\right) = \frac{\sin \omega t}{\omega}$

$\therefore L^{-1}\left[\frac{1}{s(s^2 + \omega^2)}\right] = \int_0^t \frac{\sin \omega \tau}{\omega} d\tau = \frac{1}{\omega} \left[-\frac{\cos \omega \tau}{\omega}\right]_0^t = \underline{\underline{\frac{1 - \cos \omega t}{\omega^2}}}$

ii) $\frac{2}{s(s+3)}$

Soln: Let $F(s) = \frac{1}{s+3} \Rightarrow f(t) = L^{-1}\left[\frac{1}{s+3}\right] = e^{-3t}$

$\therefore L^{-1}\left[\frac{2}{s(s+3)}\right] = 2 \int_0^t e^{-3\tau} d\tau = 2 \left[\frac{e^{-3\tau}}{-3}\right]_0^t = \underline{\underline{\frac{2}{3} [1 - e^{-3t}]}}$

H.W

i) Find $L^{-1}\left[\frac{1}{s(s+1)^3}\right]$ using integration

ii) Find $L^{-1}\left[\frac{1}{s^2(s+4)}\right]$ using integration.

(7) LT-Q1)

Solving differential equations using Laplace transform

1. Solve the initial value problem

$$y'' - y = t \quad \text{Given } y(0) = 1 \text{ and } y'(0) = 1$$

Soln:- Apply Laplace transform on both sides

$$L(y'') - L(y) = L(t)$$

$$\Rightarrow [s^2 L(y) - s y(0) - y'(0)] - L(y) = \frac{1}{s^2}$$

$$\Rightarrow (s^2 - 1) L(y) - s - 1 = \frac{1}{s^2}$$

$$\Rightarrow (s^2 - 1) L(y) = \frac{1}{s^2} + s + 1$$

$$\Rightarrow L(y) = \frac{1}{s^2(s^2 - 1)} + \frac{s}{s^2 - 1} + \frac{1}{s^2 - 1}$$

$$\Rightarrow y = L^{-1} \left[\frac{1}{s^2(s^2 - 1)} \right] + L^{-1} \left[\frac{s}{s^2 - 1} \right] + L^{-1} \left[\frac{1}{s^2 - 1} \right]$$

$$= L^{-1} \left[\frac{1}{s^2 - 1} - \frac{1}{s^2} \right] + L^{-1} \left[\frac{s}{s^2 - 1} \right] + L^{-1} \left[\frac{1}{s^2 - 1} \right]$$

$$= \sinh t - t + \cosh t + \sinh t$$

$$= 2 \sinh t + \cosh t - t$$

2. Solve $y'' - 3y' + 2y = 4t - 8$. Given $y(0) = 2$ and $y'(0) = 7$

Soln:- Apply Laplace transform on both sides.

$$L(y'') - 3L(y') + 2L(y) = 4L(t) - 8L(1)$$

$$\Rightarrow [s^2 L(y) - s y(0) - y'(0)] - 3[s L(y) - y(0)] + 2L(y) = \frac{4}{s^2} - \frac{8}{s}$$

$$\Rightarrow (s^2 - 3s + 2) L(y) - 2s - 7 + 6 = \frac{4}{s^2} - \frac{8}{s}$$

$$\Rightarrow (s^2 - 3s + 2) L(y) = \frac{4}{s^2} - \frac{8}{s} + 2s + 1$$

(c) LT- (22)

$$\Rightarrow L(y) = \frac{4 - 8s + 2s^3 + s^2}{s^2(s^2 - 3s + 2)} = \frac{4 - 8s + 2s^3 + s^2}{s^2(s-2)(s-1)}$$

$$= \frac{A}{s} + \frac{B}{s^2} + \frac{C}{s-2} + \frac{D}{s-1}$$

$$\Rightarrow 4 - 8s + 2s^3 + s^2 = As(s-2)(s-1) + B(s-2)(s-1) + Cs^2(s-1) + Ds^2(s-2)$$

Put $s=0$ $4 = 2B \Rightarrow B=2$

$s=1$ $-1 = -D \Rightarrow D=1$

$s=2$ $8 = 4C \Rightarrow C=2$

Coefficients of s^3 $2 = A + C + D \Rightarrow A = 2 - C - D = 2 - 2 - 1 = -1$

$$\therefore L(y) = \frac{-1}{s} + \frac{2}{s^2} + \frac{2}{s-2} + \frac{1}{s-1}$$

$$\therefore y = -1 + 2t + 2e^{2t} + e^t$$

3. Solve $(D^2 + 4)y = \sin 2t$ Given $y(0) = 3$, $y'(0) = 4$

Soln $L(y'') + 4L(y) = L(\sin 2t)$

$$\Rightarrow s^2 L(y) - sy(0) - y'(0) + 4L(y) = \frac{2}{s^2 + 4}$$

$$\Rightarrow (s^2 + 4)L(y) = \frac{2}{s^2 + 4} + 3s + 4$$

$$\Rightarrow L(y) = \frac{2}{(s^2 + 4)^2} + \frac{3s}{s^2 + 4} + \frac{4}{s^2 + 4}$$

$$\Rightarrow y = L^{-1}\left(\frac{2}{(s^2 + 4)^2}\right) + 3L^{-1}\left(\frac{s}{s^2 + 4}\right) + 4L^{-1}\left(\frac{1}{s^2 + 4}\right)$$

To find $L^{-1}\left(\frac{1}{(s^2 + 4)^2}\right)$ use convolution theorem.

Let $F(s) = \frac{1}{s^2 + 4}$

$$\Rightarrow f(t) = \frac{1}{2} \sin 2t \quad g(t) = \frac{1}{2} \sin 2t$$

(9) LT - (23)

$$\begin{aligned}
 L^{-1}\left(\frac{1}{(s^2+4)^2}\right) &= \int_0^t \frac{1}{2} \sin 2u \cdot \frac{1}{2} \sin 2(t-u) du \\
 &= \frac{1}{4} \int_0^t (\cos 2t + \cos(4u-2t)) du = \frac{1}{4} \int_0^t \frac{\cos(4u-2t) - \cos 2t}{2} du \\
 &= \frac{1}{4} \left[u \cos 2t + \frac{\sin(4u-2t)}{4} \right]_0^t = \frac{1}{8} \left[\frac{\sin 6t}{4} - t \cos 2t \right] \\
 &= \frac{1}{4} \left[t \cos 2t + \frac{\sin 2t}{4} - \frac{\sin 2t}{4} \right] = \frac{1}{8} \left[\frac{\sin 2t}{2} - t \cos 2t \right] \\
 &= \frac{t}{4} \cos 2t
 \end{aligned}$$

$$y = 2 \left(\frac{t}{4} \cos 2t \right) + 3 \cos 2t + \sin 2t$$

4) Solve $y' + \frac{2}{3}y = -4 \cos 2t$ Given $y(0) = 0$

Soln: $L(y') + \frac{2}{3}L(y) = -4L(\cos 2t)$

$$sL(y) - y(0) + \frac{2}{3}L(y) = \frac{-4s}{s^2+4}$$

$$\left(s + \frac{2}{3}\right)L(y) = \frac{-4s}{s^2+4}$$

$$\Rightarrow L(y) = \frac{-4s}{(s^2+4)\left(s + \frac{2}{3}\right)} = \frac{-12s}{(s^2+4)(3s+2)}$$

$$\frac{-12s}{(s^2+4)(3s+2)} = \frac{As+B}{s^2+4} + \frac{C}{3s+2}$$

$$-12s = (As+B)(3s+2) + C(s^2+4)$$

Put $s = -\frac{2}{3}$

$$8 = C\left(\frac{4}{9} + 4\right) \Rightarrow C = \frac{9}{51}$$

Coefficient of s^2 , $0 = 3A + C \Rightarrow A = \frac{-C}{3} = \frac{-9}{15}$

Coefficient of s , $-12 = 2A + 3B \Rightarrow -12 = \frac{-18}{15} + 3B \Rightarrow B = \frac{-18}{5}$

(+ve) - LT - (24)

$$y = \frac{-9}{15} L^{-1}\left(\frac{s}{s+4}\right) - \frac{18}{5} L^{-1}\left(\frac{1}{s+4}\right) + \frac{9}{15} L^{-1}\left(\frac{1}{\left(s+\frac{2}{3}\right)}\right)$$

$$= \frac{-9}{15} \cos 2t - \frac{18}{10} \sin 2t + \frac{9}{15} e^{-\frac{2t}{3}}$$

5) Solve $y'' + 2y' - 3y = 0$. Given $y(2) = -3$ & $y'(2) = -5$

Ans Here $x_0 = 2$

Introduce $\tilde{x} = x - x_0 = x - 2$

$$\therefore \tilde{y}'' + 2\tilde{y}' - 3\tilde{y} = 0 \quad \tilde{y}(0) = -3 \quad \tilde{y}'(0) = -5$$

Apply Laplace transform on both sides,

$$[s^2 L(\tilde{y}) - s\tilde{y}(0) - \tilde{y}'(0)] + 2[s L(\tilde{y}) - \tilde{y}(0)] - 3 L(\tilde{y}) = 0$$

$$\Rightarrow (s^2 + 2s - 3) L(\tilde{y}) + 3s + 5 + 6 = 0$$

$$\Rightarrow L(\tilde{y}) = \frac{-3s - 11}{(s+3)(s-1)} = \frac{1}{2(s+3)} - \frac{7}{2(s-1)} \quad (\text{Using partial fraction})$$

$$\Rightarrow \tilde{y} = L^{-1}\left[\frac{1}{2(s+3)}\right] - L^{-1}\left[\frac{7}{2(s-1)}\right] = \frac{1}{2} e^{-3\tilde{x}} - \frac{7}{2} e^{\tilde{x}}$$

$$\Rightarrow y = \frac{1}{2} e^{-3(x-2)} - \frac{7}{2} e^{x-2}$$

H.W

1) Solve $y' + 2y = 0$. Given $y(0) = 1.5$

2) Solve $y'' + y' - 6y = 0$. Given $y(0) = 1$ & $y'(0) = 1$

3) Solve $y'' + 9y = 10e^{-t}$ $y(0) = 0$ $y'(0) = 0$

4) Solve $y'' - 6y' + 5y = 29 \cos 2t$ $y(0) = 3.2$ $y'(0) = 6.2$

5) Solve $y'' + 7y' + 12y = 21e^{3t}$ $y(0) = 3.5$ $y'(0) = -10$

6) Solve $y' - 6y = 0$. Given $y(-1) = 4$

7) Solve $y'' + y = 2t$. $y\left(\frac{\pi}{4}\right) = \frac{\pi}{2}$ $y'\left(\frac{\pi}{4}\right) = 2 - \sqrt{2}$

8) Solve $y'' + 2y' + 5y = 50t - 100$. Given $y(2) = -4$ & $y'(2) = 14$

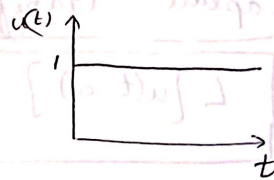
(41) LT-(25)

Unit step function and second shifting theorem

The unit step function, also called Heaviside function is defined as

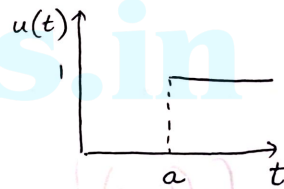
$$u(t) = \begin{cases} 0 & \text{for } t < 0 \\ 1 & \text{for } t \geq 0 \end{cases}$$

The graph will be :



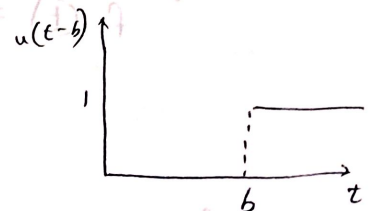
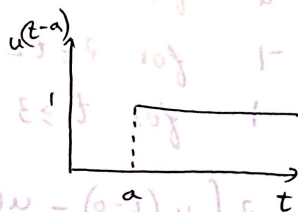
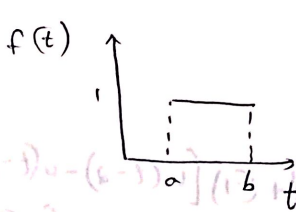
Similarly $u(t-a) = \begin{cases} 0 & \text{for } t < a \\ 1 & \text{for } t \geq a \end{cases}$

The graph will be



The fn $f(t)$ defined by $f(t) = \begin{cases} 1 & \text{for } a \leq t \leq b \\ 0 & \text{otherwise} \end{cases}$

can be expressed in terms of unit step fn as
 $f(t) = u(t-a) - u(t-b)$



(12) LT - (26)

Let $f(t)$ be a piecewise continuous fn defined as

$$f(t) = \begin{cases} f_1(t) & \text{for } 0 < t < t_1, \\ f_2(t) & \text{for } t_1 < t < t_2, \end{cases}$$

$$f(t) = f_1(t) [u(t-0) - u(t-t_1)] + f_2(t) [u(t-t_1) - u(t-t_2)]$$

Laplace transform of unit step function

$$\boxed{L[u(t-a)] = \frac{e^{-as}}{s}}$$

Proof:-

$$\begin{aligned} L[u(t-a)] &= \int_0^{\infty} e^{-st} u(t-a) dt = \int_0^a (e^{-st} \times 0) dt + \int_a^{\infty} (e^{-st} \times 1) dt \\ &= \left[\frac{e^{-st}}{-s} \right]_a^{\infty} = \frac{e^{-as}}{s} \end{aligned}$$

$$L[u(t-a)] = \frac{e^{-as}}{s} \implies$$

$$\boxed{L^{-1}\left[\frac{e^{-as}}{s}\right] = u(t-a)}$$

1. Express the function in terms of unit step function and then find its Laplace transform.

$$f(t) = \begin{cases} 2 & \text{for } 0 < t < 2 \\ -1 & \text{for } 2 \leq t < 3 \\ 1 & \text{for } t \geq 3 \end{cases}$$

$$\text{Soln :- } f(t) = 2[u(t-0) - u(t-2)] + (-1)[u(t-2) - u(t-3)] + (1)[u(t-3)]$$

$$\implies f(t) = 2 - 3u(t-2) + 2u(t-3)$$

$$L[f(t)] = L(2) - 3L[u(t-2)] + 2L[u(t-3)] = \frac{2}{s} - 3\frac{e^{-2s}}{s} + \frac{2e^{-3s}}{s}$$

(13) 17-(27)

Second Shifting Theorem

if $L[f(t)] = F(s)$, then

$$L[f(t-a)u(t-a)] = e^{-as} F(s)$$

From the above, $L^{-1}[e^{-as} F(s)] = f(t-a)u(t-a)$

i. Express the function in terms of unit step function and find its Laplace transform.

i) $g(t) = \begin{cases} 0, & t < 4 \\ (t-4)^2, & t \geq 4 \end{cases}$

Soln : $g(t) = (t-4)^2 u(t-4)$

$$\begin{aligned} L[g(t)] &= L[(t-4)^2 u(t-4)] \quad \text{Here } f(t-4) = (t-4)^2 \\ &= e^{-4s} \frac{2}{s^3} \quad (\text{by second shifting theorem}) \Rightarrow L(f(t)) = \frac{2}{s^3} = F(s) \\ &\Rightarrow f(t) = t^2 \end{aligned}$$

ii) $g(t) = t-3$ for $t \geq 3$

Soln : $g(t) = (t-3) u(t-3)$

$$\begin{aligned} L[g(t)] &= L[(t-3) u(t-3)] \quad \text{Here } f(t-3) = t-3 \\ &= e^{-3s} \frac{1}{s^2} \quad \Rightarrow f(t) = t \\ &\Rightarrow L(f(t)) = \frac{1}{s^2} = F(s) \end{aligned}$$

iii) $g(t) = t \quad (0 < t < 4)$

Soln : $g(t) = t[u(t-0) - u(t-4)]$
 $= t u(t-0) - t u(t-4)$

(44) LT - (2s)

$$\begin{aligned} L\{g(t)\} &= L\{(t-0)u(t-0)\} - L\{[(t-2)+2]u(t-2)\} \\ &= L\{(t-0)u(t-0)\} - L\{(t-2)u(t-2)\} - 2L\{u(t-2)\} \\ &= \frac{e^{-0s}}{s^2} - \frac{e^{-2s}}{s^2} - 2\frac{e^{-2s}}{s} = \frac{1}{s^2} - e^{-2s}\left[\frac{1}{s^2} + \frac{2}{s}\right] \end{aligned}$$

iv) $g(t) = e^{-t} \quad (0 < t < \pi)$

soln :- $g(t) = e^{-t} [u(t-0) - u(t-\pi)]$

$$\begin{aligned} L\{g(t)\} &= L\{e^{-(t-0)}u(t-0)\} - L\{e^{-(t-\pi)-\pi}u(t-\pi)\} \\ &= L\{e^{-(t-0)}u(t-0)\} - e^{-\pi}L\{e^{-(t-\pi)}u(t-\pi)\} \\ &= \frac{1}{s+1} - e^{-\pi}\frac{e^{-\pi s}}{s+1} \end{aligned}$$

v) $g(t) = e^{\frac{-\pi t}{2}} \quad (1 < t < 3)$

soln :- $g(t) = e^{\frac{-\pi t}{2}} [u(t-1) - u(t-3)]$

$$\begin{aligned} L\{g(t)\} &= L\left\{e^{\frac{-\pi t}{2}}u(t-1)\right\} - L\left\{e^{\frac{-\pi t}{2}}u(t-3)\right\} \\ &= L\left\{e^{\frac{-\pi}{2}(t-1)-\frac{\pi}{2}}u(t-1)\right\} - L\left\{e^{\frac{-\pi}{2}(t-3)-\frac{3\pi}{2}}u(t-3)\right\} \\ &= e^{\frac{-\pi}{2}}\frac{e^{-s}}{s+\frac{\pi}{2}} - e^{\frac{-3\pi}{2}}\frac{e^{-3s}}{s+\frac{\pi}{2}} \end{aligned}$$

vi) $g(t) = \sin t \quad \text{for } \frac{\pi}{2} < t < \pi$

soln :- $g(t) = \sin t [u(t-\frac{\pi}{2}) - u(t-\pi)]$

$$L\{g(t)\} = L\left\{\sin t u(t-\frac{\pi}{2})\right\} - L\{\sin t u(t-\pi)\}$$

(16) LT-(29)

$$= L \left\{ \sin \left(\left(t - \frac{\pi}{2} \right) + \frac{\pi}{2} \right) u \left(t - \frac{\pi}{2} \right) \right\} - L \left\{ \sin \left(\left(t - \pi \right) + \pi \right) u \left(t - \pi \right) \right\}$$

$$= L \left\{ \cos \left(t - \frac{\pi}{2} \right) u \left(t - \frac{\pi}{2} \right) \right\} + L \left\{ \sin \left(t - \pi \right) u \left(t - \pi \right) \right\}$$

$$= e^{-\frac{\pi s}{2}} \frac{s}{s^2 + 1} + \frac{e^{-\pi s}}{s^2 + 1}$$

vii) $g(t) = 2t^2 \quad \text{for } t > \frac{s}{2}$

Soln! $g(t) = 2t^2 u \left(t - \frac{s}{2} \right)$

$$L(g(t)) = L \left(2t^2 u \left(t - \frac{s}{2} \right) \right)$$

$$= 2 L \left\{ \left[\left(t - \frac{s}{2} \right)^2 + s t - \frac{s^2}{4} \right] u \left(t - \frac{s}{2} \right) \right\}$$

$$= 2 L \left\{ \left[\left(t - \frac{s}{2} \right)^2 + s \left(t - \frac{s}{2} + \frac{s}{2} \right) - \frac{s^2}{4} \right] u \left(t - \frac{s}{2} \right) \right\}$$

$$= 2 L \left\{ \left(t - \frac{s}{2} \right)^2 u \left(t - \frac{s}{2} \right) \right\} + 10 L \left\{ \left(t - \frac{s}{2} \right) u \left(t - \frac{s}{2} \right) \right\} + 2 L \left\{ \left(\frac{2s}{4} - \frac{s^2}{4} \right) u \left(t - \frac{s}{2} \right) \right\}$$

$$= 2 e^{-\frac{s s}{2}} \frac{2}{s^3} + 10 \frac{e^{-\frac{s s}{2}}}{s^2} + 2 \left(\frac{2s}{4} - \frac{s^2}{4} \right) \frac{e^{-\frac{s s}{2}}}{s}$$

$$= 2 e^{-\frac{s s}{2}} \left[\frac{2}{s^3} + \frac{s}{s^2} + \frac{2s}{4s} \right]$$

H.W

1) $g(t) = \cos 2t \quad (0 < t < \pi)$

2) $g(t) = \sin \pi t \quad (2 < t < 4)$

3) $g(t) = t^2 \quad (1 < t < 2)$

4) $g(t) = \begin{cases} 2 & 0 < t < 1 \\ \frac{t^2}{2} & 1 \leq t < \frac{\pi}{2} \\ \cos t & t \geq \frac{\pi}{2} \end{cases}$

16) $L^{-1} - (30)$

To find the inverse

1) Find the inverse of $\frac{s e^{-2s}}{s^2 + 16}$

Soln:- By second shifting theorem,

$$L^{-1} \left[e^{-as} F(s) \right] = f(t-a) u(t-a)$$

Here $a = 2$ $F(s) = \frac{s}{s^2 + 16} \Rightarrow f(t) = \cos 4t$
 $\Rightarrow f(t-2) = \cos 4(t-2)$

$$\therefore L^{-1} \left[\frac{s e^{-2s}}{s^2 + 16} \right] = \cos 4(t-2) u(t-2)$$

2) Find the inverse of $\frac{e^{-3s}}{s^6}$

Soln:- Here $a = 3$ $F(s) = \frac{1}{s^6} \Rightarrow f(t) = \frac{t^5}{5!}$

$$\Rightarrow f(t-3) = \frac{(t-3)^5}{5!}$$

$$L^{-1} \left[\frac{e^{-3s}}{s^6} \right] = \frac{(t-3)^5}{5!} u(t-3)$$

3) Find $L^{-1} \left[\frac{e^{-2s}}{(s-1)^3} \right]$

Soln:- Here $a = 2$ $F(s) = \frac{1}{(s-1)^3} \Rightarrow f(t) = \frac{t^2}{2} e^t$
 (By first shifting theorem)

$$L^{-1} \left[\frac{e^{-2s}}{(s-1)^3} \right] = f(t-2) u(t-2)$$

$$= \frac{(t-2)^2}{2} e^{t-2} u(t-2)$$

(47) LT-(311)

4) Find $L^{-1} \left[\frac{1 - e^{-\pi s}}{s^2 + 4} \right]$

Soln: $L^{-1} \left[\frac{1 - e^{-\pi s}}{s^2 + 4} \right] = L^{-1} \left[\frac{1}{s^2 + 4} \right] - L^{-1} \left[\frac{e^{-\pi s}}{s^2 + 4} \right]$

$L^{-1} \left[\frac{1}{s^2 + 4} \right] = \frac{1}{2} \sin 2t$

To find $L^{-1} \left[\frac{e^{-\pi s}}{s^2 + 4} \right]$. Here $a = \pi$
 $F(s) = \frac{1}{s^2 + 4} \Rightarrow f(t) = \frac{1}{2} \sin 2t$

$\Rightarrow f(t - \pi) = \frac{1}{2} \sin 2(t - \pi)$

$\therefore L^{-1} \left[\frac{e^{-\pi s}}{s^2 + 4} \right] = f(t - \pi) u(t - \pi) = \frac{1}{2} \sin 2(t - \pi) u(t - \pi)$

$\therefore L^{-1} \left[\frac{1 - e^{-\pi s}}{s^2 + 4} \right] = \frac{1}{2} \sin 2t - \frac{1}{2} \sin 2(t - \pi) u(t - \pi)$

H.W
 1) Find $L^{-1} \left[\frac{e^{-s}}{s^2 + \pi^2} + \frac{e^{-2s}}{s^2 + \pi^2} + \frac{e^{-3s}}{(s+2)^2} \right]$

2) Find $L^{-1} \left[\frac{e^{-2s} - 2e^{-5s}}{s} \right]$

3) Find $L^{-1} \left\{ 2 \left[\frac{e^{-s} - e^{-3s}}{s^2 - 4} \right] \right\}$

To solve differential equations

1) solve $y'' + 3y' + 2y = 1$ if $0 < t < 1$ and 0 if $t \geq 1$

Given $y(0) = 0$ $y'(0) = 0$

(48) LT - (32)

Soln:- $y'' + 3y' + 2y = u(t-0) - u(t-1)$

Apply Laplace transform on both sides,

$$L(y'') + 3L(y') + 2L(y) = L[u(t-0) - u(t-1)]$$

$$\Rightarrow [s^2 L(y) - sy(0) - y'(0)] + 3[sL(y) - y(0)] + 2L(y) = \frac{1}{s} - \frac{e^{-s}}{s}$$

$$\Rightarrow (s^2 + 3s + 2)L(y) = \frac{1}{s} - \frac{e^{-s}}{s}$$

$$\Rightarrow L(y) = \frac{1}{s(s+2)(s+1)} - \frac{e^{-s}}{s(s+1)(s+2)}$$

$$\frac{1}{s(s+2)(s+1)} = \frac{A}{s} + \frac{B}{s+2} + \frac{C}{s+1}$$

$$1 = A(s+2)(s+1) + Bs(s+1) + Cs(s+2)$$

Put $s=0 \Rightarrow 1 = 2A \Rightarrow A = \frac{1}{2}$

$s = -2 \Rightarrow 1 = 2B \Rightarrow B = \frac{1}{2}$

$s = -1 \Rightarrow 1 = -C \Rightarrow C = -1$

$$\therefore L(y) = \frac{1}{2s} + \frac{1}{2(s+2)} - \frac{1}{(s+1)} - \frac{e^{-s}}{2s} - \frac{e^{-s}}{2(s+2)} + \frac{e^{-s}}{s+1}$$

$$\Rightarrow y = \frac{1}{2} + \frac{1}{2} e^{-2t} - e^{-t} - \frac{1}{2} u(t-1) - \frac{1}{2} e^{-2(t-1)} u(t-1) + e^{-(t-1)} u(t-1)$$

HW

1) Solve $y'' + 3y' + 2y = 4t$ if $0 < t < 1$ and 8 if $t > 1$

Given $y(0) = 0$ $y'(0) = 0$

2) Solve $y'' + y = 2t$ if $0 < t < 1$ and 2 if $t > 1$

Given $y(0) = 0$ $y'(0) = 0$

3) Solve $y'' + y' - 2y = 3\sin t - \cos t$ if $0 < t < 2\pi$ and

$3\sin 2t - \cos 2t$ if $t > 2\pi$. Given $y(0) = 0$ & $y'(0) = 0$

(19) L7-(33)

a) Solve $y'' + 4y = r(t)$ where $r(t) = 4 \cos t$ if $0 < t < \pi$
and 0 if $t > \pi$

Given $y(0) = 0$ & $y'(0) = 0$

Soln : $y'' + 4y = 4 \cos t [u(t) - u(t-\pi)]$

$$\Rightarrow y'' + 4y = 4 \cos t u(t) + 4 \cos(t-\pi) u(t-\pi)$$

Apply Laplace transform on both sides

$$L(y'') + 4L(y) = L[4 \cos t u(t)] + L[4 \cos(t-\pi) u(t-\pi)]$$

$$\Rightarrow [s^2 L(y) - s y(0) - y'(0)] + 4L(y) = \frac{4s}{s^2+1} + 4 \frac{e^{-\pi s}}{s^2+1}$$

$$\Rightarrow (s^2 + 4) L(y) = \frac{4s}{s^2+1} + 4 \frac{e^{-\pi s}}{s^2+1}$$

$$\Rightarrow L(y) = \frac{4s}{(s^2+1)(s^2+4)} + 4 \frac{e^{-\pi s}}{(s^2+1)(s^2+4)}$$

$$= 4 \left[\frac{1}{3} \left(\frac{s}{s^2+1} \right) - \frac{1}{3} \left(\frac{s}{s^2+4} \right) \right] + 4 \left[\frac{e^{-\pi s}}{3(s^2+1)} - \frac{e^{-\pi s}}{3(s^2+4)} \right]$$

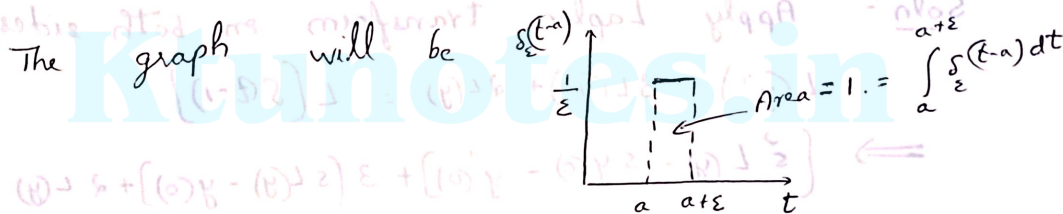
$$\Rightarrow y = \frac{4}{3} \cos t - \frac{4}{3} \cos 2t + \frac{4}{3} \cos(t-\pi) u(t-\pi) - \frac{4}{3} \cos 2(t-\pi) u(t-\pi)$$

(20) LT - (34)

Dirac Delta function or Unit impulse function

An airplane making a "hard" landing, a mechanical system being hit by a hammer blow, a ship being hit by a single high wave, a tennis ball hit by a racket and many other similar examples appear in everyday life. They are phenomena of an impulsive nature where actions & forces are applied over short interval of time. Such phenomena can be modelled by Dirac delta function.

Consider the function $\delta_\varepsilon(t-a) = \begin{cases} \frac{1}{\varepsilon} & , a < t < a+\varepsilon \\ 0 & , \text{otherwise} \end{cases}$



As $\varepsilon \rightarrow 0$, the height of the strip increases infinitely

So Dirac delta function is defined as

$$\delta(t-a) = \lim_{\varepsilon \rightarrow 0} \delta_\varepsilon(t-a) = \begin{cases} \infty & \text{for } t=a \\ 0 & \text{for } t \neq a \end{cases}$$

such that $\int_0^\infty \delta(t-a) dt = 1$

Laplace transform of $\delta(t-a)$

$$\boxed{L[\delta(t-a)] = e^{-as}}$$

Proof

$$L[\delta_\varepsilon(t-a)] = \int_a^{a+\varepsilon} e^{-st} \frac{1}{\varepsilon} dt = \frac{1}{\varepsilon} \left(\frac{e^{-st}}{-s} \right)_a^{a+\varepsilon} = \frac{1}{\varepsilon} \left[\frac{e^{-s(a+\varepsilon)}}{-s} - \frac{e^{-sa}}{-s} \right]$$

(21) LT-(35)

$$\begin{aligned}
 \text{Therefore } L\{\delta(t-a)\} &= \lim_{\varepsilon \rightarrow 0} L\left\{\delta_{\varepsilon}(t-a)\right\} \\
 &= \lim_{\varepsilon \rightarrow 0} \frac{e^{-as}}{s\varepsilon} [1 - e^{-s\varepsilon}] \\
 &= e^{-as} \lim_{\varepsilon \rightarrow 0} \left[\frac{1 - e^{-s\varepsilon}}{s\varepsilon} \right] \\
 &= e^{-as} \lim_{\varepsilon \rightarrow 0} e^{-s\varepsilon} \left[\text{Using L'Hospital's rule} \right] \\
 &= \underline{e^{-as}}
 \end{aligned}$$

i) Solve $y'' + 3y' + 2y = \delta(t-1)$. Given $y(0)=0$ & $y'(0)=0$

Soln:- Apply Laplace transform on both sides,

$$\begin{aligned}
 \Rightarrow L(y'') + 3L(y') + 2L(y) &= L\{\delta(t-1)\} \\
 \Rightarrow [s^2 L(y) - sy(0) - y'(0)] + 3[sL(y) - y(0)] + 2L(y) &= e^{-s} \\
 \Rightarrow (s^2 + 3s + 2)L(y) &= e^{-s} \\
 \Rightarrow L(y) &= \frac{e^{-s}}{(s+1)(s+2)} = \frac{e^{-s}}{s+1} - \frac{e^{-s}}{s+2} \quad (\text{using partial fractions}) \\
 \Rightarrow y &= L^{-1}\left[\frac{e^{-s}}{s+1}\right] - L^{-1}\left[\frac{e^{-s}}{s+2}\right] \\
 &= e^{-(t-1)} u(t-1) - e^{-2(t-1)} u(t-1) \\
 &= \left[e^{-(t-1)} - e^{-2(t-1)} \right] u(t-1)
 \end{aligned}$$

ii) Solve $y'' + 9y = \delta(t - \frac{\pi}{2})$ Given $y(0)=0$ & $y'(0)=0$

$$\underline{\text{Soln}}:- L(y'') + 9L(y) = L\left\{\delta\left(t - \frac{\pi}{2}\right)\right\}$$

(22) LT-(26)

$$(s^2 + 9)L(y) - s y(0) - y'(0) + 9L(y) = e^{-\frac{\pi s}{2}}$$

$$\Rightarrow (s^2 + 9)L(y) = e^{-\frac{\pi s}{2}} + 2s$$

$$\Rightarrow L(y) = \frac{e^{-\frac{\pi s}{2}}}{s^2 + 9} + \frac{2s}{s^2 + 9}$$

$$\Rightarrow y = L^{-1}\left[\frac{e^{-\frac{\pi s}{2}}}{s^2 + 9}\right] + L^{-1}\left[\frac{2s}{s^2 + 9}\right]$$

$$= \frac{1}{3} \sin 3\left(t - \frac{\pi}{2}\right) u\left(t - \frac{\pi}{2}\right) + 2 \cos 3t$$

3) Solve $y'' + 2y' + 5y = 25t - 100\delta(t - \pi)$

Given $y(0) = -2$ & $y'(0) = 5$

Soln:- $L(y'') + 2L(y') + 5L(y) = 25L(t) - 100L[\delta(t - \pi)]$

$$\Rightarrow [s^2 L(y) - s y(0) - y'(0)] + 2[s L(y) - y(0)] + 5L(y) = \frac{25}{s^2} - 100e^{-\pi s}$$

$$\Rightarrow (s^2 + 2s + 5)L(y) = \frac{25}{s^2} - 100e^{-\pi s} - 2s + 1$$

$$\Rightarrow L(y) = \frac{25}{s^2(s^2 + 2s + 5)} - 100 \frac{e^{-\pi s}}{s^2 + 2s + 5} - \frac{2s}{s^2 + 2s + 5} + \frac{1}{s^2 + 2s + 5}$$

$$= \left[\frac{-2s + 5}{s^2} + \frac{2s - 1}{s^2 + 2s + 5} \right] - 100 \frac{e^{-\pi s}}{s^2 + 2s + 5} - \frac{2s}{s^2 + 2s + 5} + \frac{1}{s^2 + 2s + 5}$$

$$= \left(-\frac{2}{s} + \frac{5}{s^2} + \frac{2(s+1)-3}{(s+1)^2 + 4} \right) - 100 \frac{e^{-\pi s}}{(s+1)^2 + 4} - \left[\frac{2(s+1)-2}{(s+1)^2 + 4} \right] + \frac{1}{(s+1)^2 + 4}$$

$$\Rightarrow y = L^{-1}\left(\frac{-2}{s}\right) + L^{-1}\left(\frac{5}{s^2}\right) + 2L^{-1}\left[\frac{(s+1)}{(s+1)^2 + 4}\right] - 3L^{-1}\left[\frac{1}{(s+1)^2 + 4}\right] - 100L^{-1}\left[\frac{e^{-\pi s}}{(s+1)^2 + 4}\right] - 2L^{-1}\left[\frac{s+1}{(s+1)^2 + 4}\right] + 2L^{-1}\left[\frac{1}{(s+1)^2 + 4}\right] + L^{-1}\left[\frac{1}{(s+1)^2 + 4}\right]$$

(23) LT - (37)

$$\therefore y = -2 + 5t + 2e^{-t} \cos 2t - \frac{3}{2} e^{-t} \sin 2t - \frac{100}{2} e^{-(t-\pi)} \sin 2(t-\pi) u(t-\pi)$$

$$- 2e^{-t} \cos 2t + e^{-t} \sin 2t + \frac{1}{2} e^{-t} \sin 2t$$

$$\Rightarrow y = -2 + 5t - 50 e^{-(t-\pi)} \sin 2(t-\pi) u(t-\pi)$$

H.W

1) Solve $y'' + 16y = 4\delta(t-3\pi)$. Given $y(0) = 2$ & $y'(0) = 0$.

2) Solve $y'' + 4y = \delta(t-\pi) - \delta(t-2\pi)$. Given $y(0) = 0$ & $y'(0) = 1$.

3) Solve $y'' + 3y' + 2y = \sin t + \delta(t-1)$. Given $y(0) = 1$ & $y'(0) = -1$.

4) Solve $y'' + 3y' + 2y = u(t-1) + \delta(t-2)$. Given $y(0) = 0$ & $y'(0) = 1$.

5) Solve $4y'' + 16y' + 17y = 3e^{-t} + \delta(t - \frac{1}{4})$. Given $y(0) = \frac{3}{5}$ & $y'(0) = \frac{3}{5}$.